

Strategic coexistence theory for evolutionary game theory

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Abstract

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin, 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis (Hofbauer and Sigmund, 1998; Cressman and Tao, 2014). However, less attention has been given to similarities between replicator dynamics and community ecology until recently (Tarnita and Traulsen, 2025; Allesina, 2026), even though the classic replicator equation stems from the generalized Lotka-Volterra equation (Page and Nowak, 2002).

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025): the dominance of a single strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a single strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory (Motro, 1991; Doebeli and Hauert, 2005; Gore et al., 2009). Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition? Even though EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanism between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms (Chesson, 2000; Godoy et al., 2014; Godoy and Levine, 2014; Kraft et al., 2015). MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to overcome average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits

its competitor (Chesson, 2000). In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions. Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each entity to compete under the limiting environments created by its competitor (Chesson, 2000). This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains (Sieben et al., 2022; Park et al., 2024, 2026). Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two strategy games, providing reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilizes competition between cooperation and defection, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies.

Strategic coexistence theory

Here, we present a brief derivation of strategic coexistence theory (SCT) (Figure 1). To do so, we begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j .

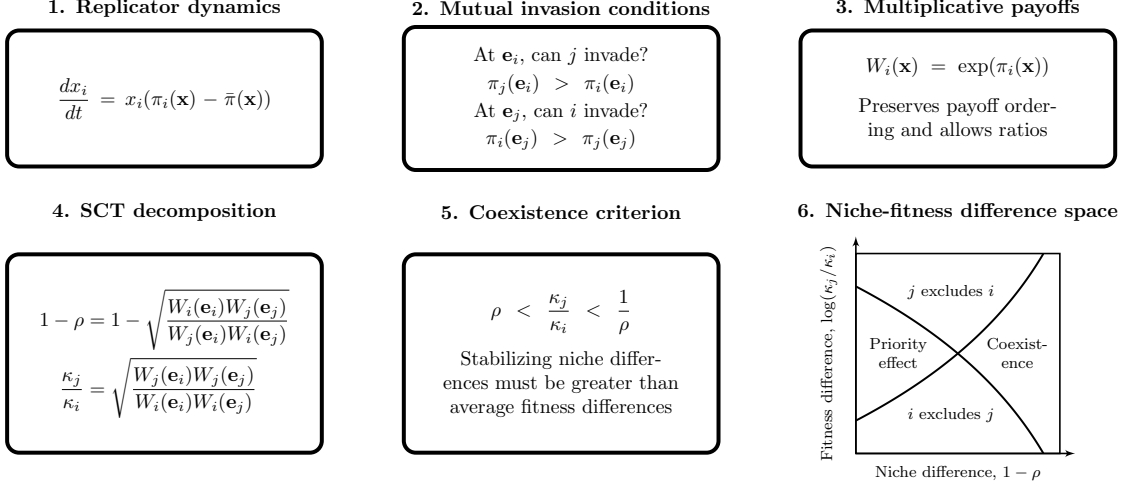


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i . Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion.

Because payoffs in many games can be negative, payoff ratios can be difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

This exponential transformation is convenient because it preserves the payoff ordering and does not alter the invasion conditions. Therefore, this allows us to compare strategic environments through ratios, as in modern coexistence theory. For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition. Second, we define the fitness difference as the ratio between the geometric means of the

93 multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

94 In other words, strategy j has an average competitive advantage over strategy i
 95 ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated
 96 by itself and its competitor.

97 Together, for any two strategies described by the classic replicator equation, mu-
 98 tual invasion requires stabilizing niche differences to overcome average fitness differ-
 99 ences. Specifically, the condition for mutual invasion can be rewritten as:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

100 This inequality applies to any games described by the classic replicator equation
 101 because ρ and κ_j/κ_i can be calculated directly from ratios of multiplicative pay-
 102 offs. This decomposition not only allows us understand how different mechanisms
 103 contribute to the coexistence or exclusion of competing strategies but also provides
 104 shared axes for comparing different games.

105 We note that the absolute magnitudes of niche and fitness differences are sensitive
 106 to payoff scaling because they are defined on an exponentiated payoff scale. However,
 107 the qualitative outcome of strategic competition under SCT is preserved because the
 108 exponential transformation does not alter invasion conditions.

109 Classic two-strategy games

110 As an illustrative example, we begin with classic two-strategy games, which can be
 111 described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

112 Here, the first and second rows represent strategies 1 and 2, respectively, and the
 113 columns represent the strategy of the interacting opponent. Writing x_1 to denote
 114 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

115 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

116 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
 117 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
 118 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
 119 the game space into four regimes (Figure 2B): prisoner's dilemma ($T > R$, $P >$
 120 S), hawk-dove game ($T > R$, $S > P$), stag-hunt game ($R > T$, $P > S$), and
 121 harmony game ($R > T$, $S > P$). In standard EGT, these outcomes are obtained by
 122 analyzing the stability of equilibrium conditions. For simple games, this analysis is
 123 straightforward.

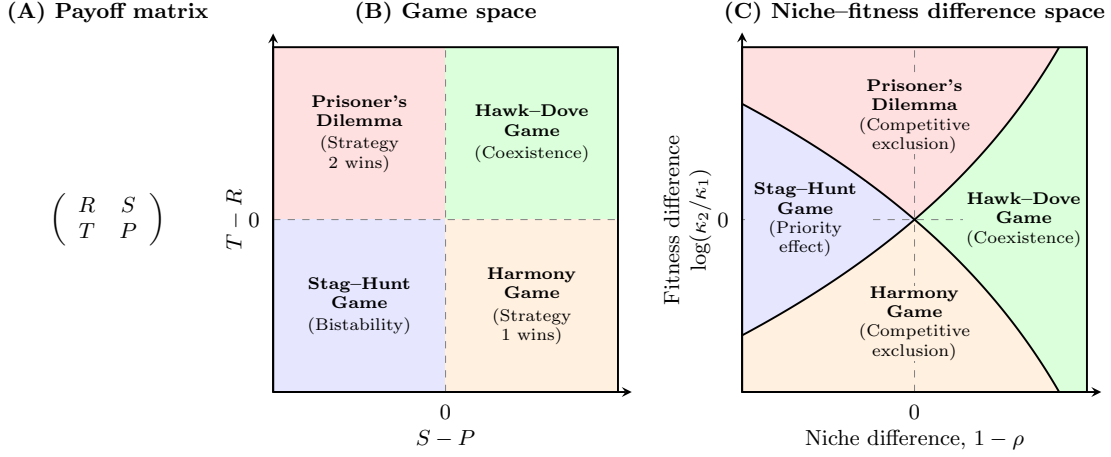


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime.

124 A coexistence-theoretic approach recovers the same classification but provides
 125 a different interpretation (Figure 2C). For a two-strategy game determined by this
 126 payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to $T > R$ and $S > P$. This corresponds to the hawk–dove game, in which each strategy can invade when rare and the two strategies coexist. The remaining games also have natural interpretations in niche–fitness difference space. Prisoner’s dilemma and harmony games represent competitive exclusion regimes, where one strategy has a fitness advantage that exceeds niche differences. In contrast, stag–hunt games represent a priority-effect regime, where neither strategy can invade the other and the outcome depends on initial conditions. Therefore, coexistence theory not only allows us to recover the classic results from two-strategy games but also provides new interpretations for these results in terms of niche and fitness differences. Building on this, we next apply this framework to comparing core mechanisms of cooperation.

Comparisons of mechanisms for the evolution of cooperation

Even though cooperative behavior can be found across a wide range of biological systems, a simple prisoner’s dilemma shows that cooperation is not an evolutionarily stable strategy. Thus, many researchers have sought to identify mechanisms that promote evolutionary stability of cooperation. Here, we compare five classic mechanisms for the evolution of cooperation using SCT and ask how they differ from the perspective of strategy coexistence.

Specifically, we consider the following mechanisms: kin selection, direct reciprocity, indirect reciprocity, network reciprocity, and group selection (Figure 3A). Kin selection favors cooperation when two interacting individuals are genetic relatives, where the genetic relatedness r must be greater than the cost-to-benefit ratio: $r > c/b$. Direct reciprocity favors cooperation when current cooperation may promote future cooperation by the same individual through repeated encounters. In this case, the probability of repeated interaction (i.e., “next round”) must be greater than the cost-to-benefit ratio: $w > c/b$. Indirect reciprocity favors cooperation when current cooperation may promote future cooperation by different individuals through reputation. Thus, the social acquaintanceship (i.e., the probability of knowing someone’s reputation) must be greater than the cost-to-benefit ratio: $q > c/b$. Network reciprocity favors cooperation via clustering of cooperators, requiring the benefit-to-cost ratio to be greater than the average number of neighbors: $b/c > k$. Finally, group selection favors cooperation via multilevel selection, where groups with more cooperators exhibit faster growth at the group level even if defectors reproduce faster at the individual level. This requires $b/c > 1 + (n/m)$, where n is the maximum group size and m is the number of groups. All of these mechanisms can be described by a simple 2×2 payoff matrix (Figure 3A), from which the corresponding niche and fitness differences can be computed using Eq. (10) and Eq. (11).

First, the prisoner’s dilemma provides a baseline scenario, where defection is evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete

(A)

Prisoner's dilemma	Kin selection
$\begin{pmatrix} b-c & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(1+r) & br-c \\ b-r & 0 \end{pmatrix}$
Direct reciprocity	Indirect reciprocity
$\begin{pmatrix} (b-c)/(1-w) & -c \\ b & 0 \end{pmatrix}$	$\begin{pmatrix} b-c & -c(1-q) \\ b(1-q) & 0 \end{pmatrix}$
Network reciprocity	Group selection
$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}$	$\begin{pmatrix} (b-c)(m+n) & (b-c)m-cn \\ bn & 0 \end{pmatrix}$

(B)

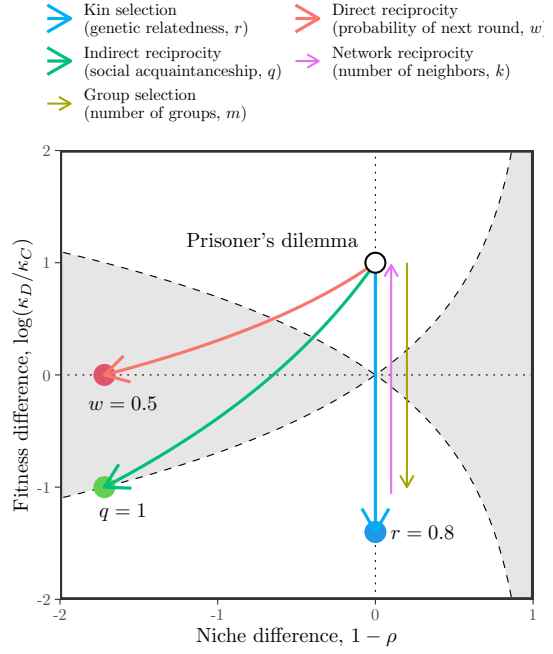


Figure 3: **Comparisons of five mechanisms for the evolution of cooperation using SCT.** (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . Network reciprocity is parameterized based on $H = [(b-c)k - 2c]/[(k+1)(k-2)]$. (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all result in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows.

167 niche overlap, and therefore defection competitively excludes cooperation (Figure
 168 3B). Building on this, SCT shows that five mechanisms promote evolutionary stabil-
 169 ity of cooperation via different dynamical routes (Figure 3B). For example, kin se-
 170 lection allows the fitness of cooperation to increase as genetic relatedness r increases
 171 (Figure 3B). However, it does not affect the niche difference between cooperation
 172 and defectors. Similarly, network reciprocity and group selection modulate the fit-
 173 ness of cooperation without causing any changes in strategic niches. In contrast,
 174 direct and indirect reciprocity destabilizes competition while increasing the fitness
 175 of cooperation, thereby pushing the competition outcome towards a priority effect
 176 regime. Equivalently, these two mechanisms promote bistability of defection and
 177 cooperation.

178 SCT further reveals another difference between direct and indirect reciprocity.
 179 In particular, direct reciprocity is unable to generate sufficient fitness difference to
 180 competitively exclude defection even when the probability of next round w reaches 1.
 181 In contrast, indirect reciprocity is able to generate sufficient fitness difference to push
 182 the system towards the boundary between priority-effect regime and competitive-
 183 exclusion regime as social acquaintanceship q reaches 1, indicating that an additional
 184 mechanism will be able to promote complete dominance of cooperation.

185 We note that none of these mechanisms allow coexistence of cooperation and
 186 defection. Therefore, we move on to the final example, applying SCT to micro-
 187 bial cooperation, where nonlinear benefits and environmental context can promote
 188 coexistence of cooperators and defectors.

189 Microbial cooperation and public goods

190 Here, we consider a simple, phenomenological model of competition between cooper-
 191 ating and cheating yeast in a cooperative sucrose-metabolism system, where invertase
 192 production by cooperators allows cheaters to utilize glucose, a public good released
 193 through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by
 194 cooperators incurs a cost c and generates a total benefit of 1 that is captured pri-
 195 vately with efficiency ϵ . Assuming a linear relationship between glucose availability
 196 and microbial growth, the payoffs for cooperators π_C and cheaters (defectors) π_D can
 197 then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

198 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 199 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

200 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 201 f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the
 202 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 203 efficiency, $c > \epsilon$ (Gore et al., 2009).

204 Our framework allows us to reinterpret this result from the perspective of coexis-
 205 tence theory. In particular, the two strategies have no niche difference, corresponding
 206 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 207 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

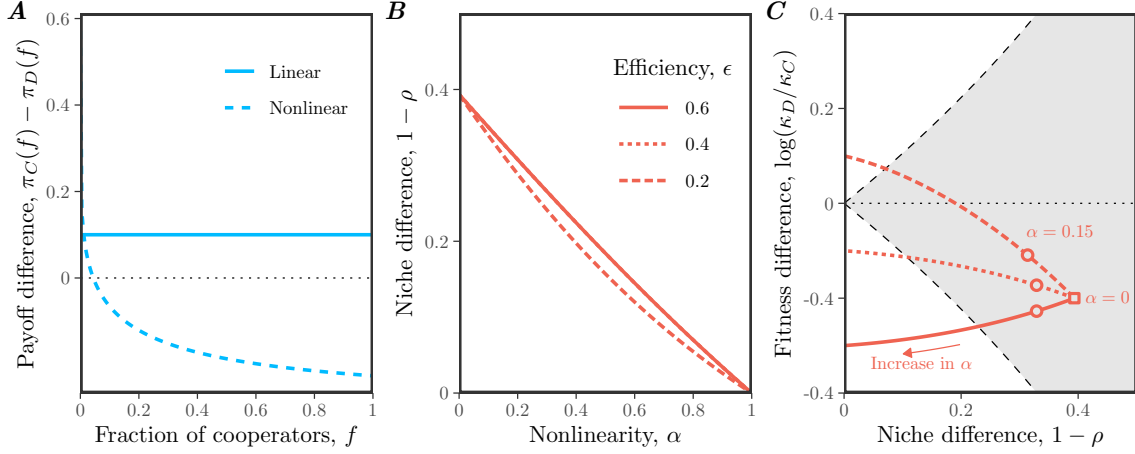


Figure 4: **Effects of nonlinear microbial growth on coexistence of cooperation and defection.** (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are completely overlapping. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha = 0$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

Therefore, this model always predicts competitive exclusion, except in the neutral case $c = \epsilon$. The outcome is determined entirely by the fitness difference between cooperators and defectors, rather than by stabilizing niche differences. This example resembles the prisoner's dilemma discussed in the previous example (Figure 3).

In real biological systems, the effect of glucose on microbial growth can follow a nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - \{f(1 - \epsilon)\}^\alpha]. \quad (19)$$

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure

219 2A): cooperators have a competitive advantage when rare, whereas defectors have a
 220 competitive advantage when cooperators are abundant.

221 From a coexistence-theoretic perspective, this means that nonlinear growth can
 222 generate sufficient stabilizing niche difference to overcome the fitness difference. In
 223 particular, we can show that the niche and fitness differences for this model are given
 224 by

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

225 These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For
 226 example, the cost of cooperation does not affect niche difference. Instead, niche
 227 difference is generated by the concave relationship between glucose availability and
 228 microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0
 229 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0^+$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

230 which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong
 231 concavity not only generates stabilizing niche differences, but also acts as an equal-
 232 izing mechanism (Figure 4C).

233 Discussion

234 Standard evolutionary game theory and replicator equations provide powerful tools
 235 for predicting the outcome of strategy competition, including strategic dominance,
 236 coexistence, and bistability. However, these outcomes are typically understood by
 237 analyzing payoff differences within individual games, which makes it difficult to com-
 238 pare coexistence mechanisms across models. Building on Chesson's Modern Coexis-
 239 tence Theory from community ecology, we developed a strategic coexistence theory
 240 (SCT) that allows us to decompose strategy competition in terms of stabilizing niche
 241 differences and fitness differences. This decomposition provides a unifying framework
 242 for comparing different games from the perspective of coexistence theory, allowing
 243 us to identify stabilizing and equalizing mechanisms for strategy competition.

244 To demonstrate the utility of SCT, we considered three case studies. First, the
 245 analysis of classic two-strategy games showed that SCT recovers the original clas-
 246 sification for these classic games, highlighting the similarities between evolutionary
 247 games and community ecology. Then, we compared different mechanisms for the evo-
 248 lution of cooperation, which showed that each mechanism promotes cooperation via
 249 different dynamical routes. Kin selection, network reciprocity, and group selection
 250 primarily affect the fitness difference between cooperation and defection, whereas

251 direct and indirect reciprocity can destabilize competition and generate priority ef-
252 fects. Finally, our analysis of microbial cooperation showed that coexistence between
253 cooperators and defectors requires more than reducing the fitness cost of coopera-
254 tion: nonlinear benefits can generate stabilizing niche differences while also reducing
255 fitness asymmetry. Taken together, these examples show that SCT can provide new
256 insights into ecological underpinnings of strategy competition.

257 There are several limitations to SCT. First, similar to MCT from community
258 ecology, SCT only allows for comparisons of pairwise strategies and therefore cannot
259 determine the outcomes of games with more than two strategies. Second, we focused
260 on classic replicator equations, but Tarnita and Traulsen (2025) recently pointed
261 out that these classic formulations implicitly ignore differences in intrinsic growth
262 rates. In such cases, replicator equations may fail to predict the outcome of com-
263 petition (Tarnita and Traulsen, 2025), and a full Lotka-Volterra model is needed to
264 describe the abundances, rather than frequencies, of strategies. Thus, SCT inherits
265 the same assumptions as the classic replicator framework: it is most appropriate
266 when strategy competition can be described in terms of relative frequencies alone.
267 When strategies differ intrinsically in growth, MCT may be required instead to tease
268 apart mechanisms of strategy coexistence. Finally, we have focused on simple exam-
269 ples to illustrate the utility of SCT, but many games involve additional complexities,
270 including population structure (Nowak et al., 2010), stochasticity (Wang et al., 2025;
271 Bodin et al., 2026), and explicit ecological feedback. Extending SCT to these settings
272 will be necessary to determine how broadly stabilizing and equalizing mechanisms
273 can be compared across evolutionary games.

274 In conclusion, SCT provides a bridge between evolutionary game theory and com-
275 munity ecology, allowing strategy competition to be understood from the perspective
276 of coexistence theory. Therefore, rather than replacing existing EGT or replicator-
277 dynamics frameworks, SCT provides a complementary tool for teasing apart the
278 coexistence mechanisms underlying evolutionary games. More broadly, our study
279 builds on recent efforts to extend MCT beyond species competition, providing a
280 foundation for building unifying coexistence theory for competing entities, including
281 species, pathogen strains, and behavioral strategies.

Supplementary Text

S1 Derivation of the strategic coexistence theory

To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

where x_i represents the frequency of strategy i , π_i represents the payoff of strategy i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

we obtain the following inequality that defines conditions for mutual invasion and long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

295 S2 Applications to classic two strategy games

296 Consider a two strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

297 There are four classic games that we can define from this simple payoff matrix:
 298 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 299 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 300 each game corresponds to a distinct region in the niche-fitness difference space under
 301 SCT.

302 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 303 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 304 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

305 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

306 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

307 As before, SCT allows us to predict the mutual invasion of two strategy using the
 308 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

309 Then, simple algebra reveals that each condition in mutual invasion inequality
 310 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

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